

HAMEDAN, Iran, Islamic Republic ofWMO: **407680**Lat: **34.8688N**Lon: **48.5344E**Elev: **1749**StdP: **82.00**Time Zone: **3.50 (IRN)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.1	(c) -13.0	(d) -19.3	(e) 0.8	(f) -12.1	(g) -16.4	(h) 1.1	(i) -6.7	(j) 10.6	(k) 6.9	(l) 9.6	(m) 5.5	(n) 0.3	(o) 120	(p) 0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.3	(c) 36.2	(d) 16.1	(e) 35.1	(f) 15.8	(g) 34.0	(h) 15.5	(i) 18.5	(j) 32.2	(k) 17.3	(l) 31.5	(m) 16.4	(n) 31.2	(o) 3.8	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	11.7	25.6	11.6	10.6	24.4	10.3	9.6	22.5	60.1	32.2	55.7	31.4	52.4	31.1	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.3	7.1	-19.1	38.2	4.1	0.9	-22.1	38.8	-24.5	39.3	-26.9	39.9	-29.9	40.5		
			-19.3	19.4	4.2	1.8	-22.3	20.7	-24.7	21.8	-27.1	22.8	-30.2	24.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	-1.6	1.2	6.9	11.3	16.1	21.6	25.2	24.6	19.6	13.7	6.3	1.7		
	DBStd	9.66	5.91	4.94	4.01	3.28	2.60	2.33	1.96	1.74	2.27	3.01	3.63	4.26		
	HDD10.0	1123	361	246	112	23	0	0	0	0	0	6	118	257		
	HDD18.3	2778	619	479	355	211	79	3	0	0	13	144	361	514		
	CDD10.0	1957	0	1	16	62	188	348	472	452	289	122	8	1		
	CDD18.3	570	0	0	0	1	8	101	214	193	51	2	0	0		
	CDH23.3	9317	0	0	1	16	268	1655	3128	2948	1192	109	0	0		
	CDH26.7	4702	0	0	0	1	43	736	1788	1661	466	7	0	0		
(14) Wind	WSAvg	2.3	1.8	2.4	3.2	3.1	2.7	2.3	2.5	2.3	2.0	2.0	1.6	1.6		
(15) Precipitation	PrecAvg	332	36	39	56	55	27	4	4	1	3	23	45	40		
	PrecMax	513	60	81	107	107	70	21	22	10	38	82	214	132		
	PrecMin	256	9	6	12	11	3	0	0	0	0	0	3	1		
	PrecStd	59	14	19	22	26	19	5	6	2	8	25	41	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.6	22.3	25.3	29.8	35.8	38.2	37.1	33.5	27.4	19.9	16.0		
		MCWB	4.4	6.4	8.6	11.7	13.8	15.6	16.4	16.4	13.8	11.4	8.3	6.1		
	2%	DB	10.9	12.9	19.8	22.7	27.9	33.9	36.9	36.1	32.1	26.0	17.4	13.0		
		MCWB	3.8	5.2	7.6	10.7	12.9	15.5	16.2	16.1	13.6	11.2	7.7	5.5		
	5%	DB	8.8	10.9	17.2	20.8	26.2	32.2	35.8	35.0	31.0	24.5	15.3	10.8		
		MCWB	3.0	4.3	6.8	9.9	12.3	15.0	16.0	15.6	13.3	10.7	7.0	4.4		
	10%	DB	6.8	8.9	15.1	19.1	24.7	30.9	34.2	34.0	29.8	22.9	13.4	8.8		
		MCWB	2.0	3.5	6.2	9.0	11.8	14.7	15.9	15.2	12.9	10.2	6.5	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.6	9.7	12.7	15.7	18.3	20.1	20.3	16.1	13.3	10.2	7.6		
		MCDB	10.5	13.0	17.5	21.0	25.4	30.7	34.4	33.5	29.6	24.1	15.5	13.6		
	2%	WB	4.6	6.1	8.5	11.6	14.2	17.0	18.6	18.6	14.9	12.0	9.2	6.2		
		MCDB	8.7	11.1	17.5	20.3	24.6	30.3	32.7	32.2	29.1	22.8	14.7	11.4		
	5%	WB	3.5	4.8	7.5	10.7	13.4	16.1	17.6	16.8	14.0	11.3	8.1	5.0		
		MCDB	7.9	9.6	15.7	18.8	23.7	29.6	32.0	31.4	28.9	22.1	13.5	9.2		
	10%	WB	2.2	3.8	6.6	9.9	12.5	15.3	16.7	15.8	13.2	10.6	7.2	3.9		
		MCDB	6.1	8.4	13.9	17.2	22.5	29.1	31.7	31.9	28.0	21.0	12.3	7.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.2	12.6	12.9	15.4	18.1	18.3	19.2	19.7	16.4	12.0	11.0		
		MCDBR	13.4	14.2	15.7	16.0	18.1	19.9	19.8	20.2	20.8	19.0	15.3	14.6		
		MCWBR	7.9	7.9	7.5	6.9	7.4	8.3	7.7	7.8	8.6	8.7	7.9	8.3		
		MCDBR	11.9	12.3	13.5	14.0	16.4	18.6	18.2	19.0	19.6	16.6	12.6	12.1		
	5% WB	MCWBR	7.3	7.2	7.0	6.4	7.1	8.2	7.7	7.9	8.5	7.9	7.0	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.332	0.385	0.463	0.458	0.433	0.473	0.428	0.365	0.380	0.327	0.294		
	taud		2.370	2.257	2.098	1.915	1.957	2.032	1.929	2.053	2.225	2.151	2.321	2.402		
	Ebn at Noon		911	906	881	826	835	852	815	848	890	836	859	888		
	Edh at Noon		93	118	151	191	185	172	189	164	131	129	98	85		
(44) All-Sky Solar Radiation	RadAvg		2.92	3.89	4.83	5.72	6.92	8.19	7.73	7.23	6.22	4.49	3.10	2.52		
	RadStd		0.24	0.35	0.42	0.49	0.50	0.36	0.29	0.24	0.23	0.31	0.31	0.30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.06	-1.31	-1.51	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.36	N/A	N/A	+0.73	-0.83	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page